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RESEARCH PROJECT REPORT

Characterization of the Dirichlet-to-Neumann Operator for the Wave Equation

AUTHOR **Massaro Elias**

SUPERVISOR M. Dos Santos Ferreira

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1. Introduction

1.1 The Dirichlet-to-Neumann operator

Let $\Omega \subset \mathbb{R}^n$ be an open domain with sufficiently regular boundary $\Gamma = \partial\Omega$. Consider a partial differential equation posed on Ω , endowed with a Dirichlet condition on Γ . The Dirichlet-to-Neumann operator (hereafter denoted DtN) is the map that assigns to any boundary datum f the normal flux of the corresponding solution.

Definition 1.1 (Dirichlet-to-Neumann operator). *Let u be the solution of the PDE under consideration satisfying $u|_{\Gamma} = f$. The DtN operator is defined by*

$$\Lambda : f|_{\Gamma} \longmapsto \partial_{\nu} u|_{\Gamma},$$

where ∂_{ν} denotes the outward normal derivative on Γ .

1.2 Motivation

The DtN operator arises naturally in several contexts in analysis, mathematical physics and scientific computing.

In *electrical impedance tomography*, one seeks to reconstruct the conductivity $\sigma(x)$ inside a body from measurements of currents and voltages performed on the boundary. Mathematically, this amounts to determining whether knowledge of the map Λ entirely characterizes the interior differential operator: this is the *Calderón problem* [1].

In *numerical simulation*, the DtN operator provides transparent boundary conditions, which make it possible to truncate an unbounded domain while avoiding artificial reflections. Knowledge of an explicit and numerically stable expression of Λ is then essential.

1.3 Introductory example: the Laplacian

Before turning to the wave equation, let us recall the classical result for the Laplace equation on the half-space $\mathbb{R}^{n-1} \times \mathbb{R}_+^*$. We consider the problem

$$\begin{cases} -\Delta u = 0, & x_n > 0, \\ u(x', 0) = f(x'), \end{cases}$$

together with the decay condition $u \rightarrow 0$ as $x_n \rightarrow +\infty$. Applying the Fourier transform in the tangential variables $x' \in \mathbb{R}^{n-1}$, the equation reduces to an ODE in x_n :

$$(-|\xi'|^2 + \partial_{x_n}^2) \hat{u}(\xi', x_n) = 0.$$

The bounded solution is $\hat{u}(\xi', x_n) = \hat{f}(\xi') e^{-|\xi'|x_n}$, whence

$$\widehat{\Lambda f}(\xi') = -\partial_{x_n} \hat{u}|_{x_n=0} = |\xi'| \hat{f}(\xi').$$

The DtN operator for the Laplacian is therefore the pseudodifferential operator with symbol $|\xi'|$, that is, $\Lambda = \sqrt{-\Delta_{x'}}$. It is an operator of order 1.

1.4 Conventions and definitions

This section collects the notions that occur repeatedly in the report. They are recalled here once and for all.

Symbol of an operator. To any operator A acting on functions on $\mathbb{R}^d$ one associates, whenever this makes sense, its symbol $\sigma_A(\xi)$: the function of the dual variable such that $\widehat{Af}(\xi) = \sigma_A(\xi) \hat{f}(\xi)$ on the Fourier side. For a differential operator with constant coefficients, the symbol is polynomial (for instance $\sigma_{-\Delta}(\xi) = |\xi|^2$). For the operators studied here it will be non-polynomial, whence the name pseudodifferential. When a Laplace transform in time is involved, the symbol depends both on the dual variable ξ' and on the Laplace variable s : one then speaks of a Fourier-Laplace symbol.

Order of an operator. The order of an operator A is the integer (or real number) m such that its symbol satisfies $|\sigma_A(\xi)| \sim C |\xi|^m$ as $|\xi| \rightarrow \infty$. For a differential operator, the order coincides with the degree of the highest derivative: ∇ is of order 1, Δ of order 2. For a pseudodifferential operator, m may be any real number.

Causality. A phenomenon is said to be causal if the state of the system at a time t depends only on earlier times. For the wave equation, this property is stronger: the solution at (x, t) depends only on the content of the domain of dependence $\{(y, \tau) : \tau \leq t, |x - y| \leq t - \tau\}$, since waves propagate at finite speed (here normalized to 1). Symmetrically, a datum injected at a point influences only the future light cone emanating from that point.

Throughout the report we assume that the boundary datum f vanishes for $t \leq 0$ and that the initial conditions $u|_{t=0} = \partial_t u|_{t=0} = 0$ are satisfied. The solution is then causal as well: the DtN kernels computed below will systematically be supported in cones of the type $\{|x'| < t\}$ or $\{r < t - \tau\}$.

These vanishing initial conditions also have a useful computational consequence. The identity $\mathcal{L}\{f''(t)\}(s) = s^2 \hat{f}(s) - s f(0) - f'(0)$ reduces to $\mathcal{L}\{f''\}(s) = s^2 \hat{f}(s)$. This simplification makes legitimate, on the disc (Chapter 3) and on the cylinder (Chapter 4), the derivative transfer that consists in rewriting $s^2 \hat{f}$ as the transform of f'' in order to make the spectral series absolutely convergent.

Hadamard finite part. When an integral $\int_{\mathbb{R}^d} \varphi(x)/r^k dx$ diverges at the origin ($r = |x|$, with k too large for the dimension d), one gives it a distributional meaning through the Hadamard finite part, denoted Pf. One truncates the integral to $r > \varepsilon$, subtracts the explicitly identified terms diverging in ε , and passes to the limit $\varepsilon \rightarrow 0^+$. In dimension 2,

$$\langle \text{Pf } r^{-3}, \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \left(\int_{r > \varepsilon} \frac{\varphi(x)}{r^3} dx - \frac{2\pi \varphi(0)}{\varepsilon} \right).$$

The finite part appears in Chapter 2 in order to regularize the DtN kernel on the light cone.

Fundamental solution. A *fundamental solution* of a linear differential operator L is a distribution E satisfying

$$L E = \delta_0,$$

where δ_0 is the Dirac mass at the origin (in space-time: $\delta_0 = \delta(x)\delta(t)$). Intuitively, E is the response of L to a single pointwise impulsive excitation. Knowledge of E makes it possible to solve $Lu = S$ for an arbitrary right-hand side S by a simple convolution, $u = E * S$. We shall retain in particular the retarded fundamental solution of the wave operator $\square = \partial_t^2 - \Delta$ in dimension three,

$$E(x, t) = \frac{\delta(t - |x|)}{4\pi |x|},$$

which will be used in Chapter 2 in the treatment of the half-space perturbed by a potential.

1.5 Approach and outline

The guiding principle of the study has been the treatment of the DtN operator by integral transforms: Fourier in the tangential variables, Laplace (or Fourier) in time, then inversion. This approach reduces the PDE to a parametrized ordinary differential equation, whose resolution yields the symbol of Λ . The difficulty is then shifted onto the inversion of this symbol back to physical space, and it is at this stage that the main obstructions appeared. A recurring observation of the project is that the difficulty lies in the transform chosen to invert it rather than in the problem itself: depending on the strategy adopted, one does or does not run into non-summable singularities. The project was therefore conducted in an exploratory manner, several methods having been tried, abandoned, and then resumed from another angle. The report keeps a trace of this process, in particular in Chapter 2, where two distinct computations of the half-space kernel are presented side by side without yet being completely reconciled. A formal framework exists for these phenomena – the theory of Lagrangian distributions associated with a pair of cleanly intersecting Lagrangians [2] – but assimilating it exceeded the time allotted to the project; more elementary tools were substituted for it (explicit transforms, Hadamard finite part, Bessel functions, factorizations).

The report is organized in four parts. Chapter 2 deals with the DtN operator on the **half-space**, first unperturbed through two independent computations (Bessel/Hankel tables; distributional Laplacian), then perturbed by a potential via the Born approximation. Chapter 3 addresses the **unit disc**, where the bounded geometry imposes an inversion strategy based on the Hadamard-Weierstrass factorization. Chapter 4 moves the problem onto the **finite hollow cylinder**, where the symbol becomes matrix-valued and the inversion relies on a Mittag-Leffler expansion; this part is completed by a detailed numerical validation. Chapter 5 finally addresses the **inverse problem** on the disc and on the cylinder.

2. The DtN operator on the half-space

2.1 Unperturbed DtN: two computations of the kernel

Statement of the problem

We consider the homogeneous wave equation on the half-space $\Omega = \mathbb{R}^{n-1} \times \mathbb{R}_+^*$, endowed with a Dirichlet condition on the boundary $\Gamma = \{x_n = 0\}$ and with vanishing initial conditions:

$$\begin{cases} (\partial_t^2 - \Delta_{x'} - \partial_{x_n}^2) u(x', x_n, t) = 0, & x_n > 0, t > 0, \\ u(x', 0, t) = f(x', t), & \text{(Dirichlet)} \\ u(x', x_n, 0) = \partial_t u(x', x_n, 0) = 0, & \text{(initial conditions)} \\ u(x', x_n, t) \rightarrow 0 & \text{as } x_n \rightarrow +\infty. \end{cases} \quad (2.1)$$

The boundary datum f is assumed to be regular and to vanish for $t \leq 0$, which ensures the causality of the solution. The decay condition at infinity selects the physical solution among the solutions of the ODE obtained after transformation. The associated Dirichlet-to-Neumann operator is

$$\Lambda_0 f = -\partial_{x_n} u|_{x_n=0+},$$

the sign $-$ reflecting the choice of the outward normal of the half-space, $\nu = -e_{x_n}$.

Symbol

The symbol of Λ_0 is obtained by reducing (2.1) to an ODE in x_n parametrized by the tangential dual variable and the Laplace variable in time. Applying the Fourier transform in $x' \in \mathbb{R}^{n-1}$ (dual variable ξ') and the Laplace transform in $t > 0$ (variable s , $\text{Re } s > 0$), the PDE becomes

$$(\partial_{x_n}^2 - (s^2 + |\xi'|^2)) \hat{u}(\xi', x_n, s) = 0. \quad (2.2)$$

The general solution is a combination of $e^{\pm x_n \sqrt{s^2 + |\xi'|^2}}$; the decay condition at infinity retains only the decreasing exponential, and the Dirichlet condition fixes the constant. One thus obtains

$$\hat{u}(\xi', x_n, s) = \hat{f}(\xi', s) e^{-x_n \sqrt{s^2 + |\xi'|^2}},$$

and then, evaluating $\widehat{\Lambda_0 f} = -\partial_{x_n} \hat{u}|_{x_n=0}$,

$$\widehat{\Lambda_0 f}(\xi', s) = \sqrt{s^2 + |\xi'|^2} \hat{f}(\xi', s). \quad (2.3)$$

The delicate question is the *inversion* of (2.3) back to physical space (x', t) . In accordance with the exploratory approach set out in §1.5, two lines of computation were followed, which produce two expressions for the same kernel. We present them in turn, in the chronological order of the project: first the route of the *mid-term defence* (Bessel/Hankel tables and finite part), then the route of the *distributional Laplacian*.

2.1.1 First computation: Bessel–Hankel tables

We split the symbol into a local part and a non-local part:

$$\widehat{\Lambda}f(\xi', s) = \hat{f}\left(\sqrt{s^2 + |\xi'|^2} - s\right) + \hat{f}s. \quad (2.4)$$

The term $\hat{f}s$ inverts immediately into $\partial_t f$ (local part). The corrective term $\sqrt{s^2 + |\xi'|^2} - s$ is the non-local kernel. We seek

$$K_{\text{nonlocal}}(x', t) = \mathcal{L}^{-1} \mathcal{F}^{-1} \left[\sqrt{s^2 + |\xi'|^2} - s \right].$$

Definition: Hankel transform. The Hankel transform of order ν of a function $g : \mathbb{R}_+ \rightarrow \mathbb{C}$ is defined by

$$\mathcal{H}_\nu[g](\rho) = \int_0^{+\infty} g(r) J_\nu(\rho r) r \, dr.$$

For a radial function $h(x') = g(|x'|)$ on $\mathbb{R}^2$, passing to polar coordinates together with the identity $\int_0^{2\pi} e^{-i\rho r \cos \varphi} d\varphi = 2\pi J_0(\rho r)$ gives $\mathcal{F}[h](\xi') = 2\pi \mathcal{H}_0[g](|\xi'|)$. Inverting the Fourier transform of a radial function therefore amounts to inverting a Hankel transform of order 0.

Laplace inversion followed by Hankel inversion. The table of transforms [4, 5] gives, for the Laplace inversion in time,

$$\mathcal{L}^{-1} \left[\sqrt{s^2 + |\xi'|^2} - s \right](t) = \frac{\rho}{t} J_1(\rho t), \quad \rho := |\xi'|,$$

where J_1 is the Bessel function of the first kind of order 1. The expression depends on ξ' only through its modulus ρ , so that the Fourier inversion in dimension $n - 1 = 2$ reduces to a Hankel transform of order 0. The table provides

$$\mathcal{F}^{-1} \left[\frac{\rho}{t} J_1(\rho t) \right](x') = \frac{H(t - |x'|) t}{(t^2 - |x'|^2)^{3/2}},$$

where H is the Heaviside function, which embodies causality: the kernel is supported in the cone $|x'| < t$. One obtains

$$K_{\text{nonlocal}}(x', t) = \frac{H(t - |x'|) t}{(t^2 - |x'|^2)^{3/2}}, \quad K_{\text{local}}(x', t) = -\delta'(t) \delta(x'). \quad (2.5)$$

Singularity and finite part. Writing $\Lambda f = K_{\text{local}} * f + K_{\text{nonlocal}} * f$ and letting the local term act,

$$\Lambda f(x', t) = \partial_t f + \iint_{|x' - y'| < t - \tau} \frac{(t - \tau)}{((t - \tau)^2 - |x' - y'|^2)^{3/2}} f(y', \tau) dy' d\tau. \quad (2.6)$$

The integrand diverges like $d^{-3/2}$ at the boundary of the cone, with $d := (t - \tau)^2 - |x' - y'|^2 \rightarrow 0$. The integral therefore does not exist in the ordinary sense, but only in the sense of the Hadamard finite part. An integration by parts in τ combined with the regularization lowers the order of the singularity from $d^{-3/2}$ (non-integrable) to $d^{-1/2}$ (integrable), at the cost of transferring the

derivative onto the datum:

$$\Lambda f(x', t) = \partial_t f + \iint_{\text{cone}} \frac{1}{\sqrt{(t-\tau)^2 - r^2}} \partial_\tau f(y', \tau) dy' d\tau, \quad r = |x' - y'|. \quad (2.7)$$

One further checks that K_{nonlocal} satisfies, away from the singularity ($r \neq t, t > 0$), the Poisson–Euler–Darboux equation

$$\partial_t^2 K_{\text{nonlocal}} + \frac{3}{t} \partial_t K_{\text{nonlocal}} - \Delta_{x'} K_{\text{nonlocal}} = 0,$$

which constitutes an internal verification of the computation.

Order of the operator. The symbol $\sqrt{s^2 + |\xi'|^2}$ is homogeneous of degree 1: replacing (s, ξ') by $(\lambda s, \lambda \xi')$ multiplies it by λ . The DtN operator of the wave equation is therefore of order 1, like the local part ∂_t . For the non-local part, the asymptotic expansion $J_1(z) \sim \sqrt{2/\pi z} \cos(z - 3\pi/4)$ inserted into $\hat{K}_{\text{nonlocal}}(\xi', t) = (\rho/t) J_1(\rho t)$ gives

$$\hat{K}_{\text{nonlocal}}(\xi', t) \propto \frac{\sqrt{|\xi'|}}{t^{3/2}} e^{i|\xi'|t} \quad (|\xi'| \rightarrow \infty),$$

that is, an order $1/2$ in space and $-3/2$ in time. The passage to the order in temporal frequency is carried out by Fourier transform in t : an algebraic decay in $t^{-\alpha}$ as $t \rightarrow \infty$ corresponds, by the generalized Riemann–Lebesgue lemma, to a symbol in $|\omega|^{\alpha-1}$ as $|\omega| \rightarrow \infty$, hence here an order $-1/2$ in temporal frequency. The sum equals 0, strictly less than the order 1 of ∂_t : the non-local kernel does not raise the global order.

2.1.2 Second computation: distributional Laplacian

The second route, subsequent to the defence, avoids tables of transforms and computes the kernel in the sense of distributions. It rests on the formal identity

$$\sqrt{s^2 + |\xi'|^2} = (s^2 + |\xi'|^2) \cdot \frac{1}{\sqrt{s^2 + |\xi'|^2}} \xrightarrow{\mathcal{F}^{-1}} (s^2 - \Delta_{x'}) \left(\frac{e^{-sr}}{2\pi r} \right), \quad r = |x'|, \quad (2.8)$$

where $e^{-sr}/(2\pi r)$ is the inverse Fourier transform in dimension 2 of $1/\sqrt{s^2 + |\xi'|^2}$. The difficulty is that e^{-sr}/r has a $1/r$ singularity at the origin: integrable in dimension 2 (hence a well-defined distribution), but whose classical Laplacian on $\{r > 0\}$ does not coincide with the distributional Laplacian on the whole of $\mathbb{R}^2$. Identity (2.8) must therefore be read with $\Delta_{x'}$ in the sense of distributions.

Distributional Laplacian of e^{-sr}/r . For $r > 0$, in polar coordinates ($\Delta = \partial_r^2 + r^{-1}\partial_r$), the direct computation gives

$$\Delta \left(\frac{e^{-sr}}{r} \right) \Big|_{r>0} = e^{-sr} \left(\frac{s^2}{r} + \frac{s}{r^2} + \frac{1}{r^3} \right),$$

an expression valid pointwise on $\mathbb{R}^2 \setminus \{0\}$ but not locally integrable. The computation in the sense of distributions, by truncation to $\{r > \varepsilon\}$ and Green’s formula, produces a boundary term $2\pi \varphi(0)/\varepsilon$ which cancels exactly against the divergence of the volume term in $1/r^3$, provided one defines the

Hadamard finite part

$$\langle \text{Pf } r^{-3}, \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \left(\int_{r > \varepsilon} \frac{\varphi(x')}{r^3} dx' - \frac{2\pi \varphi(0)}{\varepsilon} \right).$$

One obtains the distributional identity, valid on the whole of $\mathbb{R}^2$,

$$\Delta \left(\frac{e^{-sr}}{r} \right) = \frac{s^2 e^{-sr}}{r} + \frac{s e^{-sr}}{r^2} + e^{-sr} \text{Pf } r^{-3}. \quad (2.9)$$

Remark 2.1 (No Dirac mass). *In contrast with the three-dimensional case, where $\Delta(1/r) = -4\pi \delta_0$, no Dirac mass appears here: the fundamental solution of the Laplacian in dimension 2 is $\frac{1}{2\pi} \ln r$ and not $1/r$, so that $1/r$ is not meant to produce a Dirac mass. The singularity $1/r^3$ is entirely absorbed by the finite part.*

Kernel in the Laplace domain. Substituting (2.9) into (2.8), the terms in $s^2 e^{-sr}/r$ cancel exactly and there remains

$$K(x', s) := \mathcal{F}_{\xi'}^{-1} \left[\sqrt{s^2 + |\xi'|^2} \right] = -\frac{e^{-sr}}{2\pi} \left[\frac{s}{r^2} + \text{Pf } r^{-3} \right]. \quad (2.10)$$

Laplace inversion and final form. With $\mathcal{L}^{-1}[e^{-sr}](t) = \delta(t - r)$ and $\mathcal{L}^{-1}[s e^{-sr}](t) = \delta'(t - r)$, one obtains the space-time kernel

$$K(x', t) = -\frac{1}{2\pi} \left[\frac{\delta'(t - r)}{r^2} + \delta(t - r) \text{Pf } r^{-3} \right],$$

and then, since $\Lambda f = K *_{(x', t)} f$, letting the temporal Dirac distributions act ($\int \delta(t - \tau - r) f d\tau = f(y', t - r)$ and $\int \delta'(t - \tau - r) f d\tau = \partial_t f(y', t - r)$),

$$\boxed{\Lambda f(x', t) = -\frac{1}{2\pi} \text{Pf} \int_{\mathbb{R}^2} \left[\frac{f(y', t - r)}{r^3} + \frac{\partial_t f(y', t - r)}{r^2} \right] dy', \quad r = |x' - y'|,} \quad (2.11)$$

the finite part bearing *jointly* on the combination of the two terms, the integral being in fact restricted to the causal disc $\{r < t\}$.

Compensation mechanism. A Taylor expansion of f and $\partial_t f$ in a neighbourhood of $y' = x'$ (where $r \rightarrow 0^+$) clarifies the meaning of the symbol $\text{Pf} f$: the leading singularity of the integrand is $f(x', t)/r^3$, non-integrable in dimension 2, and it is precisely this term that requires the counter-term $2\pi f(x', t)/\varepsilon$. The contributions in $\partial_t f/r^2$ (coming from the Taylor expansions) arising from the two terms cancel exactly, and the remainder, in $\partial_t^2 f/r$, is locally integrable in polar coordinates. The regularization therefore acts on the *combination*:

$$\text{Pf} \int_{\mathbb{R}^2} \left[\frac{f(y', t - r)}{r^3} + \frac{\partial_t f(y', t - r)}{r^2} \right] dy' := \lim_{\varepsilon \rightarrow 0^+} \left(\int_{\varepsilon < r < t} \left[\frac{f(y', t - r)}{r^3} + \frac{\partial_t f(y', t - r)}{r^2} \right] dy' - \frac{2\pi f(x', t)}{\varepsilon} \right).$$

2.1.3 Comparison of the two computations

The two preceding computations produce two expressions for the same operator Λ_0 , whose explicit identification could not be carried through at the time of writing. The term involving the time derivative is explicit in the first computation and appears, through a Taylor expansion, in the

second without being isolated there. The intuition comes from the comparison with the disc (Chapter 3): there one sees a term $\sum_n |n| f_n e^{in\theta} = \sqrt{-\Delta_\theta} f$ appear, the analogue of the symbol $|\xi'|$ of the Laplacian. By consistency, one expects a contribution of type $\sqrt{-\Delta_{x'}}$ on the half-space as well: it would correspond, in (2.11), to the distributional term f/r^3 regularized by Pf, which is indeed, up to a constant, the kernel of $\sqrt{-\Delta_{x'}}$ in dimension 2. The chain of Laplace and Hankel tables of the first computation manipulates singularities at the limit of validity of the tables and would deserve to be redone in the sense of distributions, in the spirit of the second computation; completing this comparison is a direct perspective.

2.2 DtN perturbed by a potential

We now have at our disposal the elements needed to treat the case where the wave equation is perturbed by a potential. The principle is to write the perturbed problem as an integral equation, and then to truncate the iteration at first order (Born approximation) in order to obtain an explicit expression of the new DtN operator in terms of the potential.

2.2.1 Statement of the problem

We resume the setting (2.1) in dimension three, adding a potential $q \in C_c^\infty(\Omega)$ to the left-hand side of the PDE, which becomes $(\partial_t^2 - \Delta_{x'} - \partial_{x_3}^2 + q(x))u = 0$. The Dirichlet and initial conditions are unchanged. The perturbed DtN operator is defined by $\Lambda_q f = -\partial_{x_3} u|_{x_3=0}$.

2.2.2 Integral formulation and Born approximation

We rewrite the perturbed problem in the form $\square u = -q u =: S(x, t)$, where $\square = \partial_t^2 - \Delta$ is the d'Alembert operator and S is treated as a source term. The Green's function G of the half-space, constructed by the method of images, is by definition the object that inverts $\square$ while respecting the homogeneous Dirichlet condition on $\{x_3 = 0\}$:

$$\square_{x,t} G(x, y, t - \tau) = \delta(x - y) \delta(t - \tau), \quad G|_{x_3=0} = 0, \quad G(x, y, t) = E(x - y, t) - E(x - y^*, t),$$

with $y^* = (y', -y_3)$. By linearity of $\square$, one superposes the elementary responses: each source point $S(y, \tau)$ produces at (x, t) the contribution $G(x, y, t - \tau) S(y, \tau)$, and their sum provides a particular solution $u_{\text{part}} = G * S$. Indeed, applying $\square_{x,t}$ under the integral, the sifting properties of the two Dirac distributions give back exactly $\square_{x,t} u_{\text{part}} = S$. The general solution is obtained by adding the solution of the unperturbed homogeneous problem ($q = 0$), denoted u_0 , which alone carries the Dirichlet datum f since the integral term vanishes on the boundary by construction of G . Reinserting the definition of the source $S = -q u$, whose sign is taken out of the integral, one arrives at the Lippmann–Schwinger integral equation [8]:

$$u(x, t) = u_0(x, t) - \int G(x, y, t - \tau) q(y) u(y, \tau) dy d\tau. \quad (2.12)$$

This equation is exact but implicit, the unknown u appearing also under the integral through the source $S = -q u$, which motivates the Born approximation. The *first-order Born approximation* consists in replacing u by u_0 on the right-hand side:

$$u(x, t) \approx u_0(x, t) - \int G(x, y, t - \tau) q(y) u_0(y, \tau) dy d\tau. \quad (2.13)$$

This approximation is valid when **the potential q is sufficiently small in norm.**

2.2.3 Change of variables

Replacing $G = E - E^*$ in (2.13) and expanding, the corrective term splits according to the two terms of the image Green's function:

$$I_1(x, t) := \iint E(x-y, t-\tau) q(y) u_0(y, \tau) dy d\tau, \quad I_2(x, t) := - \iint E(x-y^*, t-\tau) q(y) u_0(y, \tau) dy d\tau,$$

so that the sign -1 affecting E^* in $G = E - E^*$ is *incorporated into I_2* . With these conventions, the expansion of (2.13) reads exactly

$$u(x, t) \approx u_0(x, t) - I_1(x, t) - I_2(x, t),$$

the two minus signs coming respectively from the global sign of the Born iterate.

Contribution I_1 . We introduce the relative change of variables $z = x - y$ (Jacobian 1), which makes the distance $|x - y| = |z|$ *independent* of the observation point x . The integral becomes

$$I_1(x, t) = \iint E(z, t - \tau) q(x - z) u_0(x - z, \tau) dz d\tau.$$

Inserting $E(z, t - \tau) = \delta(t - \tau - |z|)/(4\pi|z|)$, the Dirac distribution selects $\tau = t - |z|$ and the integral in τ disappears:

$$I_1(x, t) = \int_{D_1(x)} F(x, z) dz, \quad F(x, z) = \frac{q(x - z) u_0(x - z, t - |z|)}{4\pi|z|}.$$

The support in z is the portion of the past cone $|z| \leq t$; combined with the constraint $y_3 > 0$, which translates here into $x_3 - z_3 > 0$, one obtains

$$D_1(x) = \{z \in \mathbb{R}^3 : |z| \leq t, z_3 < x_3\}.$$

Contribution I_2 . For I_2 , we perform the change of variables adapted to the mirror point, $z = x - y^*$ with $y^* = (y', -y_3)$. It follows that $y = (x' - z', z_3 - x_3)$ (Jacobian 1), and $E(x - y^*) = E(z)$. The constraint $y_3 > 0$ becomes $z_3 - x_3 > 0$, that is, $z_3 > x_3$. The same mechanism of integration in τ against the Dirac distribution gives

$$I_2(x, t) = - \int_{D_2(x)} G(x, z) dz, \quad D_2(x) = \{z \in \mathbb{R}^3 : |z| \leq t, z_3 > x_3\},$$

where the integrand, *a priori distinct* from F since it is evaluated at the mirrored point, equals

$$G(x, z) = \frac{q(x' - z', z_3 - x_3) u_0(x' - z', z_3 - x_3, t - |z|)}{4\pi|z|}.$$

The minus sign in front of the integral is that of $-E^*$, in accordance with the definition of I_2 . We therefore finally have

$$u(x, t) \approx u_0(x, t) - \int_{D_1(x)} F(x, z) dz + \int_{D_2(x)} G(x, z) dz. \quad (2.14)$$

2.2.4 Lemma: normal differentiation and the Leibniz rule

The evaluation of $\Lambda_q f = -\partial_{x_3} u|_{x_3=0^+}$ requires differentiating I_1 and I_2 with respect to x_3 . However, the domains $D_1(x)$ and $D_2(x)$ depend on x_3 , which forbids naively exchanging differentiation and integration. The Leibniz rule for an integral with a variable limit reads, for a variable *upper* limit $a(x_3)$,

$$\partial_{x_3} \int_{\mathbb{R}^2} \int_{-\infty}^{a(x_3)} H(x, z) dz_3 dz' = a'(x_3) \int_{\mathbb{R}^2} H(x, z', a(x_3)) dz' + \int_{\mathbb{R}^2} \int_{-\infty}^{a(x_3)} \partial_{x_3} H(x, z) dz_3 dz'. \quad (2.15)$$

For I_1 , the domain $\{z_3 < x_3\}$ has a variable upper limit $a(x_3) = x_3$, $a'(x_3) = +1$:

$$\partial_{x_3} I_1 = \int_{\mathbb{R}^2} F(x, z', x_3) dz' + \int_{D_1(x)} \partial_{x_3} F(x, z) dz. \quad (2.16)$$

For I_2 , the domain $\{z_3 > x_3\}$ has a variable *lower* limit; using $\partial_{x_3} \int_{x_3}^{+\infty} g dz_3 = -g(x_3) + \int_{x_3}^{+\infty} \partial_{x_3} g dz_3$ and taking into account the minus sign in the definition of I_2 ,

$$\partial_{x_3} I_2 = - \left[- \int_{\mathbb{R}^2} G(x, z', x_3) dz' + \int_{D_2(x)} \partial_{x_3} G(x, z) dz \right] = \int_{\mathbb{R}^2} G(x, z', x_3) dz' - \int_{D_2(x)} \partial_{x_3} G(x, z) dz. \quad (2.17)$$

For the explicit computation of $\partial_{x_3} F$, the distance $|z|$ has become a dummy integration variable, hence $\partial_{x_3}(|z|^{-1}) = 0$: the derivative acts only on the numerator. By the chain rule and $\partial_{x_3}(x - z) = e_3$,

$$\partial_{x_3} F(x, z) = \frac{1}{4\pi|z|} \left[(\partial_3 q)(x - z) u_0(x - z, t - |z|) + q(x - z) (\partial_3 u_0)(x - z, t - |z|) \right]. \quad (2.18)$$

Remark 2.2. The invariance $\partial_{x_3}(|z|^{-1}) = 0$ is precisely the point of the relative change of variables performed in §2.2.

2.2.5 Expression of the perturbed DtN operator

We evaluate $\Lambda_q f = -\partial_{x_3} u|_{x_3=0^+}$ by combining (2.14), (2.16) and (2.17). Starting from $u = u_0 - I_1 - I_2$, we have $-\partial_{x_3} u = -\partial_{x_3} u_0 + \partial_{x_3} I_1 + \partial_{x_3} I_2$, whence

$$\begin{aligned} \Lambda_q f(x', t) &= -\partial_{x_3} u_0|_{0^+} + \partial_{x_3} I_1|_{0^+} + \partial_{x_3} I_2|_{0^+} \\ &= \Lambda_0 f(x', t) + \lim_{x_3 \rightarrow 0^+} \left[\underbrace{\int_{\mathbb{R}^2} F(x, z', x_3) dz'}_{\text{boundary term of } I_1} + \int_{D_1} \partial_{x_3} F dz \right. \\ &\quad \left. + \underbrace{\int_{\mathbb{R}^2} G(x, z', x_3) dz'}_{\text{boundary term of } I_2} - \int_{D_2} \partial_{x_3} G dz \right]. \end{aligned}$$

We analyse separately the boundary terms and the volume terms in the limit $x_3 \rightarrow 0^+$ (where $D_1 \rightarrow \{z_3 < 0\}$ and $D_2 \rightarrow \{z_3 > 0\}$). *Boundary terms.* At $z_3 = x_3$, since q has compact support the boundary terms vanish exactly

$$\int_{\mathbb{R}^2} F(x, z', x_3) dz' = \int_{\mathbb{R}^2} G(x, z', x_3) dz' = 0.$$

Volume terms. There remains $\int_{D_1} \partial_{x_3} F dz - \int_{D_2} \partial_{x_3} G dz$, that is, in the limit, $\int_{\{z_3 < 0\}} \partial_{x_3} F dz -$

$\int_{\{z_3 > 0\}} \partial_{x_3} G \, dz$. We evaluate the two integrands at $x_3 = 0$. For the first, $\partial_{x_3} F|_{x_3=0}(x, z) =: \Phi(z)$. For the second, G depends on x_3 only through the mirrored argument $z_3 - x_3$, and $\partial_{x_3}(z_3 - x_3) = -1$ introduces a sign

$$\partial_{x_3} G|_{x_3=0}(x, z) = - \frac{(\partial_3 q)(x' - z', z_3) u_0(x' - z', z_3, t - |z|) + q(x' - z', z_3) (\partial_3 u_0)(x' - z', z_3, t - |z|)}{4\pi|z|} =: -\Psi(z).$$

The second integral therefore equals $-\int_{\{z_3 > 0\}} \partial_{x_3} G \, dz = +\int_{\{z_3 > 0\}} \Psi(z) \, dz$. We perform in it the reflection $w = (z', -z_3)$ (Jacobian 1, $|z|$ invariant), which maps $\{z_3 > 0\}$ onto $\{z_3 < 0\}$; a direct computation shows that $\Psi(z)|_{z_3=-w_3} = \Phi(w)$ exactly. Hence

$$-\int_{\{z_3 > 0\}} \partial_{x_3} G(x, z) \, dz = \int_{\{w_3 < 0\}} \Phi(w) \, dw = \int_{D_0} \partial_{x_3} F(x, z) \, dz, \quad D_0 = \{z \in \mathbb{R}^3 : |z| \leq t, z_3 < 0\},$$

and the two volume terms add up to $2 \int_{D_0} \partial_{x_3} F(x, z) \, dz$. We thus obtain

$$\boxed{\Lambda_q f(x', t) = \Lambda_0 f(x', t) + 2 \int_{D_0} \partial_{x_3} F(x, z) \, dz}, \quad (2.19)$$

with

$$F(x, z) = \frac{q(x - z) u_0(x - z, t - |z|)}{4\pi|z|}, \quad D_0 = \{z \in \mathbb{R}^3 : |z| \leq t, z_3 < 0\}.$$

2.2.6 Explicit form

Inserting (2.18) into the integral over D_0 and setting $X = (x' - z', -z_3) \in \mathbb{R}^2 \times \mathbb{R}_+^*$ (with $-z_3 > 0$ since $z_3 < 0$ on D_0) and $T = t - |z|$ (retarded time), we obtain

$$\Lambda_q f(x', t) \approx \Lambda_0 f(x', t) + \frac{1}{2\pi} \int_{D_0} \frac{(\partial_3 q)(X) u_0(X, T) + q(X) (\partial_3 u_0)(X, T)}{|z|} \, dz, \quad (2.20)$$

where $(\partial_3 q)(X)$ (resp. $(\partial_3 u_0)(X, T)$) denotes the partial derivative of q (resp. u_0) with respect to its third space variable, evaluated at the point X .

2.2.7 Poisson formula for the interior field

To close the system, it remains to express the interior field u_0 in the volume D_0 *solely* in terms of the boundary datum f . We apply the space-time *Green identity* to $u = u_0$ (satisfying $\square u_0 = 0$) and $v = G$, the Green's function of the half-space introduced in (2.12) (satisfying $\square G = \delta_{(y,0)}$ and $G|_{x_3=0} = 0$). The volume terms reduce to $-u_0(y, T)$ upon integration against the Dirac mass, the initial terms disappear thanks to the vanishing initial conditions, and on the boundary $\{x_3 = 0\}$ only the contribution $u_0 \partial_\nu G$ survives since G vanishes there. The explicit computation of $\partial_\nu G|_{x_3=0}$, using the expression of E in dimension three, leads, after a classical computation, to the *Poisson kernel* of the wave equation. The retarded Dirac distribution $\delta(t - r)$ constrains the integration to the characteristic disc $\{|z'| \leq \sqrt{t^2 - x_3^2}\}$ (empty as long as $t < x_3$), and one obtains

$$u_0(x, t) = \frac{x_3}{2\pi} \int_{|z'| \leq \sqrt{t^2 - x_3^2}} \left[\frac{f(x' - z', t - r)}{r^3} + \frac{\partial_t f(x' - z', t - r)}{r^2} \right] \, dz', \quad r = \sqrt{|z'|^2 + x_3^2}. \quad (2.21)$$

Inserting (2.21) into the integral over D_0 in (2.20) yields the complete first Born iterate, expressing $\Lambda_q f$ entirely in terms of f and q .

2.2.8 Consistency of the Poisson formula with the DtN kernel

The Poisson formula (2.21) has a structure close to that of the half-space DtN kernel obtained in §2.1, which provides a cross-check of the two computations. It is written, up to a geometric factor $x_3/(2\pi)$, as an integral of the binomial $f/r^3 + \partial_t f/r^2$, which is exactly that of the distributional DtN kernel (2.11). This is no coincidence: the Kirchhoff formula and the Poisson formula both derive from the same fundamental solution $E = \delta(t - |x|)/(4\pi |x|)$, and the DtN kernel is nothing but the trace of the normal derivative of this representation.

Differentiating (2.21) with respect to x_3 and then taking the trace $x_3 \rightarrow 0^+$, the product rule produces two contributions: the one where the derivative acts on the factor x_3 (which simply becomes $1/(2\pi)$) and the one where it acts on the integral (multiplied by the prefactor x_3 , which is crushed at the trace). The mechanism is that of the second computation (§2.1.2): the singularity $1/r^3$ at $z' = 0$ is compensated by the counter-term Pf. Up to the sign of the outward normal derivative, one recovers exactly formula (2.11).

No finite part is needed in the formula for u_0 As long as $x_3 > 0$. In (2.21) one has $r = \sqrt{|z'|^2 + x_3^2} \geq x_3 > 0$: the kernels $1/r^3$ and $1/r^2$ are bounded and the integral converges in the ordinary sense. The factor x_3 plays the role of a regularizer. The finite part becomes necessary only when passing to the trace $x_3 \rightarrow 0^+$, where r may vanish at $z' = 0$; this is the role of the counter-term $2\pi f(x', t)/\varepsilon$ identified in §2.1.2, which therefore appears *only* in the DtN kernel (2.11), never in the Poisson formula itself.

3. The DtN operator on the unit disc

The preceding chapter dealt with an unbounded domain. We now turn to a bounded domain: the unit disc of $\mathbb{R}^2$.

3.1 Statement of the problem

We consider the wave equation in the unit disc $\Omega = \{(r, \theta) : r < 1\}$ of $\mathbb{R}^2$, endowed with the Dirichlet condition on the circle $\Gamma = \{r = 1\}$. The unperturbed DtN operator is

$$\Lambda f = \partial_r u|_{r=1}.$$

On the disc, the outward unit normal at the boundary is $\nu = +e_r$, so that $\partial_\nu u = \partial_r u$: the absence of a minus sign in front of ∂_r is consistent with the general definition $\Lambda f = \partial_\nu u|_\Gamma$ given in Chapter 2, where $-\partial_{x_n}$ appeared instead because the outward normal of the half-space $\{x_n > 0\}$ is $-e_{x_n}$.

The boundary datum is decomposed into an angular Fourier series

$$f(\theta, t) = \sum_{n \in \mathbb{Z}} f_n(t) e^{in\theta},$$

with the vanishing initial conditions $f_n(0) = f'_n(0) = 0$ for every $n \in \mathbb{Z}$.

3.2 Modal resolution and symbol

For each mode n , we seek u in the form $u(r, \theta, t) = R_n(r, s) e^{in\theta}$ in the Laplace domain. The wave equation in polar coordinates leads to the modified Bessel equation in r

$$r^2 R_n'' + r R_n' - (s^2 r^2 + n^2) R_n = 0.$$

The solution bounded at $r = 0$ is $R_n(r, s) = \hat{f}_n(s) I_n(sr) / I_n(s)$, where I_n is the modified Bessel function of the first kind.

Proposition 3.1 (Modal symbol). *The symbol of the DtN operator for the mode n on the unit disc is*

$$\widehat{\Lambda} f_n(s) = s \frac{I'_n(s)}{I_n(s)} \hat{f}_n(s). \quad (3.1)$$

Proof. One computes directly $\partial_r R_n|_{r=1} = \hat{f}_n(s) s I'_n(s) / I_n(s)$. □

3.3 Hadamard-Weierstrass factorization

In order to invert the symbol (3.1) in the time domain, we use the Hadamard-Weierstrass factorization theorem [3] applied to the logarithmic derivative $\frac{d}{ds} \ln I_n(s)$.

The relation $I_n(s) = i^{-n} J_n(is)$ makes it possible to locate the zeros of I_n . Since the positive real zeros $(j_{n,k})_{k \geq 1}$ of J_n are the only zeros of J_n on the positive real axis, one deduces that I_n vanishes exactly at the points $s = \pm i j_{n,k}$, $k \in \mathbb{N}^*$, located on the *purely imaginary axis*. The modal symbol $\sigma_n(s) = s I'_n(s) / I_n(s)$ therefore admits *poles* at these same points (since I_n appears there in the

denominator). Moreover, for integer n one has $I_{-n} = I_n$, so that the symbol depends only on $|n|$; we shall therefore replace n by $|n|$ in what follows.

The Hadamard-Weierstrass theorem [3] applies to any entire function h of finite order: the logarithmic derivative h'/h admits a decomposition into simple poles, up to a polynomial term that can often be shown to vanish (parity, controlled growth). For functions that are even (or odd) in s , the zeros group into pairs ζ_k and $-\zeta_k$, and the decomposition reads $h'/h = p/s + \sum_k 2s/(s^2 - \zeta_k^2)$ if h vanishes to order p at 0.

Applied to $h = I_{|n|}$, which vanishes to order $|n|$ at $s = 0$ (since $I_{|n|}(s) \sim (s/2)^{|n|}/|n|!$ near zero) and whose other zeros are $\pm i j_{|n|,k}$, this theorem gives, after multiplication by s ,

$$s \frac{I'_n(s)}{I_n(s)} = |n| + \sum_{k=1}^{\infty} \frac{2s^2}{s^2 + j_{|n|,k}^2}. \quad (3.2)$$

The polynomial term allowed *a priori* by the finite order of $I_{|n|}$ is here identically zero.

Remark 3.2. *The static limit $s \rightarrow 0$ of the symbol equals $|n|$, which coincides with the DtN operator of the Laplacian on the disc.*

The poles $\pm i j_{|n|,k}$ correspond to the Dirichlet eigenfrequencies of the disc, that is, to the values $\omega = j_{|n|,k}$ for which there exists a non-trivial free oscillation $u(x) e^{-i\omega t}$ satisfying $-\Delta u = \omega^2 u$ in Ω with $u = 0$ on $\partial\Omega$. For the unit disc, these resonances are precisely the zeros of the Bessel functions $J_{|n|}$. At these frequencies, the DtN symbol diverges.

3.4 Divergence problem and regularization

The direct term-by-term inversion of (3.2) in the time domain — which was the first strategy attempted — reveals an obstruction. The inverse Laplace transform of each fraction $s^2/(s^2 + j_{n,k}^2)$ indeed reads

$$\mathcal{L}^{-1} \left[\frac{s^2}{s^2 + j_{n,k}^2} \right] (t) = \mathcal{L}^{-1} \left[1 - \frac{j_{n,k}^2}{s^2 + j_{n,k}^2} \right] (t) = \delta(t) - j_{n,k} \sin(j_{n,k} t),$$

and the summation $\sum_k \delta(t)$ diverges in the space of distributions, while $\sum_k j_{n,k} \sin(j_{n,k} t)$ converges only in the sense of tempered distributions and remains highly singular.

To circumvent this problem, we *transfer the derivative* onto the datum. We rewrite the product in the Laplace domain by isolating the singular numerator s^2 :

$$\widehat{\Lambda f}_n(s) = |n| \hat{f}_n(s) + \sum_{k=1}^{\infty} \frac{2}{s^2 + j_{n,k}^2} \cdot [s^2 \hat{f}_n(s)]. \quad (3.3)$$

Now, thanks to the vanishing initial conditions $f_n(0) = f'_n(0) = 0$, we have the fundamental identity $s^2 \hat{f}_n(s) = \mathcal{L}\{f''_n(t)\}(s)$, which then yields, mode by mode, a convolution with a bounded kernel $\sin(j_{n,k} t)/j_{n,k}$. This transfer strategy will be reused in Chapter 4 for the cylinder.

Moreover, the inverse Laplace transform of the resolvent term is immediate:

$$\mathcal{L}^{-1} \left[\frac{2}{s^2 + j_{n,k}^2} \right] (t) = \frac{2 \sin(j_{n,k} t)}{j_{n,k}}.$$

3.5 Final expression in the time domain

Applying the inverse Laplace transform to (3.3), the frequency-domain product $\frac{2}{s^2 + j_{n,k}^2} \cdot s^2 \hat{f}_n(s)$ turns into a temporal convolution:

$$\mathcal{L}^{-1} \left[\frac{2}{s^2 + j_{n,k}^2} \cdot s^2 \hat{f}_n(s) \right] (t) = \int_0^t \frac{2 \sin(j_{n,k}(t - \tau))}{j_{n,k}} f_n''(\tau) d\tau.$$

We therefore obtain the final expression, mode by mode,

$$[\Lambda f]_n(t) = |n| f_n(t) + \sum_{k=1}^{\infty} \int_0^t \frac{2 \sin(j_{n,k}(t - \tau))}{j_{n,k}} f_n''(\tau) d\tau. \quad (3.4)$$

Summing over all azimuthal modes, and with $f(\theta, t) = \sum_n f_n(t) e^{in\theta}$, the DtN operator on the unit disc reads globally

$$\Lambda f(\theta, t) = \sum_{n \in \mathbb{Z}} \left\{ |n| f_n(t) + \sum_{k=1}^{\infty} \int_0^t \frac{2 \sin(j_{|n|,k}(t - \tau))}{j_{|n|,k}} f_n''(\tau) d\tau \right\} e^{in\theta}. \quad (3.5)$$

The term $\sum_n |n| f_n e^{in\theta}$ is recognized as $\sqrt{-\Delta_\theta} f$ (the 1D analogue of the symbol $|\xi'|$ of the Laplacian). The second term involves kernels oscillating at the frequencies $j_{n,k}$. These $j_{n,k}$ are the eigenfrequencies of vibration of a circular membrane clamped at its boundary.

Expression (3.4) will serve as the basis for the reconstruction of the radius treated in Chapter 5.

4. The DtN operator on the finite hollow cylinder

4.1 Statement of the problem

We consider the propagation of waves on the lateral surface of a finite hollow cylinder of radius $R > 0$ and length $L > 0$ (a tube open at both ends), parametrized by

$$(\theta, z) \in [0, 2\pi) \times (0, L).$$

This surface is intrinsically *flat* (vanishing Gaussian curvature; it can be unrolled onto a rectangle), which translates into a metric $ds^2 = R^2 d\theta^2 + dz^2$ and a Laplacian with constant coefficients

$$\Delta_\Omega = \frac{1}{R^2} \partial_\theta^2 + \partial_z^2.$$

The boundary $\partial\Omega$ consists of two disjoint circles:

- the lower boundary $\Gamma_0 = \{(\theta, 0) \mid \theta \in [0, 2\pi)\}$;
- the upper boundary $\Gamma_L = \{(\theta, L) \mid \theta \in [0, 2\pi)\}$.

The homogeneous wave equation on this surface reads

$$(\partial_t^2 - \Delta_\Omega)u(\theta, z, t) = 0 \quad \text{in } \Omega \times \mathbb{R}_+. \quad (4.1)$$

We impose Dirichlet conditions on the two boundary components

$$\begin{cases} u(\theta, 0, t) = f_0(\theta, t), \\ u(\theta, L, t) = f_L(\theta, t), \end{cases} \quad (4.2)$$

together with the vanishing initial conditions $u(\cdot, 0) = \partial_t u(\cdot, 0) = 0$.

In contrast with the two preceding examples, the boundary here consists of *two* connected components. The DtN operator therefore becomes *matrix-valued*: it assigns to the pair of data a pair of normal fluxes,

$$\Lambda \begin{pmatrix} f_0 \\ f_L \end{pmatrix} = \begin{pmatrix} -\partial_z u|_{z=0} \\ +\partial_z u|_{z=L} \end{pmatrix}. \quad (4.3)$$

The different sign on each boundary reflects the choice of the outward normal.

4.2 Matrix symbol via Fourier-Laplace

To analyse this operator, we use the Laplace transform in time (variable s , $\text{Re}(s) > 0$) and the Fourier series decomposition in θ (azimuthal mode $m \in \mathbb{Z}$). Applying these transformations to the wave equation gives, for each mode m , the parametrized ODE

$$(\partial_z^2 - \mu_m(s)^2)\hat{u}_m(z) = 0, \quad \mu_m(s) = \sqrt{s^2 + \frac{m^2}{R^2}}. \quad (4.4)$$

The solution adapted to the two Dirichlet conditions is

$$\hat{u}_m(z, s) = \hat{f}_{0,m}(s) \frac{\sinh(\mu_m(L - z))}{\sinh(\mu_m L)} + \hat{f}_{L,m}(s) \frac{\sinh(\mu_m z)}{\sinh(\mu_m L)}, \quad (4.5)$$

which, upon differentiation and evaluation at the boundaries, gives the matrix symbol

$$\hat{\Lambda}_m(s) = \begin{pmatrix} \mu_m \coth(\mu_m L) & -\frac{\mu_m}{\sinh(\mu_m L)} \\ -\frac{\mu_m}{\sinh(\mu_m L)} & \mu_m \coth(\mu_m L) \end{pmatrix}. \quad (4.6)$$

The diagonal term reflects the return to a given boundary of the wave emitted by that same boundary, while the off-diagonal term measures the coupling between the two boundaries (the wave emitted by Γ_0 that reaches Γ_L , and conversely).

4.3 Passage to the time domain

4.3.1 Mittag-Leffler expansion

The Laplace inversion of (4.6) rests on two classical expansions, which we briefly derive here from the Hadamard infinite product for $\sinh$.

Infinite product for $\sinh$. Recall that an entire function h is said to be *of order* ρ if ρ is the smallest real number such that $|h(z)| \leq \exp(|z|^{\rho+\varepsilon})$ for every $\varepsilon > 0$ and $|z|$ large enough; intuitively, the order measures the growth rate of h at infinity, and determines the form of the Hadamard infinite product.

The function $z \mapsto \sinh(z)$ is entire, of order 1, and vanishes exactly at the points $z = ik\pi$ for $k \in \mathbb{Z}$ (simple zeros). The Hadamard factorization theorem [3] allows one to write

$$\sinh(z) = z \prod_{k=1}^{\infty} \left(1 + \frac{z^2}{k^2 \pi^2} \right). \quad (4.7)$$

The factors $(1 + z^2/k^2 \pi^2)$ group together the pairs of conjugate zeros $\pm ik\pi$, and the prefactor z accounts for the zero at the origin. Convergence of the product is ensured by $\sum 1/k^2 < \infty$.

Expansion of $\coth$. Taking the logarithmic derivative of (4.7), one obtains

$$\coth(z) = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{2z}{z^2 + k^2 \pi^2}. \quad (4.8)$$

Expansion of $1/\sinh$. The function $1/\sinh(z)$ is meromorphic on $\mathbb{C}$ with simple poles at the points $z = ik\pi$, $k \in \mathbb{Z}$. The residue at $z = ik\pi$ is computed directly:

$$\text{Res}_{z=ik\pi} \left(\frac{1}{\sinh(z)} \right) = \frac{1}{\cosh(ik\pi)} = \frac{1}{\cos(k\pi)} = (-1)^k.$$

The Mittag-Leffler theorem then allows one to write

$$\frac{1}{\sinh(z)} = \sum_{k \in \mathbb{Z}} \frac{(-1)^k}{z - ik\pi}.$$

Grouping the terms k and $-k$,

$$\frac{(-1)^k}{z - ik\pi} + \frac{(-1)^{-k}}{z + ik\pi} = \frac{2(-1)^k z}{z^2 + k^2\pi^2},$$

one arrives at the partial fraction expansion

$$\frac{1}{\sinh(z)} = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{2(-1)^k z}{z^2 + k^2\pi^2}. \quad (4.9)$$

Applied to $z = \mu_m L$, these two expansions yield, after rearrangement,

$$\mu_m \coth(\mu_m L) = \frac{1}{L} + \frac{2}{L} \sum_{k=1}^{\infty} \frac{s^2 + \alpha}{s^2 + \omega_{m,k}^2}, \quad (4.10)$$

$$\frac{\mu_m}{\sinh(\mu_m L)} = \frac{1}{L} + \frac{2}{L} \sum_{k=1}^{\infty} \frac{(-1)^k (s^2 + \alpha)}{s^2 + \omega_{m,k}^2}, \quad (4.11)$$

where we have set $\alpha = (m/R)^2$ and $\omega_{m,k}^2 = (k\pi/L)^2 + \alpha$. The eigenfrequencies $\pm i\omega_{m,k}$ are the physical poles of the system.

4.3.2 Laplace inversion by derivative transfer

The direct Laplace inversion of the resolvent kernel $2/(s^2 + \omega^2)$ gives $2\sin(\omega t)/\omega$. The factor $s^2 + \alpha$ in the numerator, on the other hand, is problematic: its direct inversion produces Dirac distributions and their derivatives, and the infinite summation diverges in the space of distributions.

As for the disc, we transfer the derivative onto the datum. Writing

$$\frac{s^2 + \alpha}{s^2 + \omega_{m,k}^2} \hat{f}(s) = \frac{1}{s^2 + \omega_{m,k}^2} \cdot [(s^2 + \alpha)\hat{f}(s)],$$

and exploiting the vanishing initial conditions $f(0) = f'(0) = 0$, which give $\mathcal{L}\{f''\}(s) = s^2 \hat{f}(s)$, one obtains the convolution

$$\mathcal{L}^{-1} \left[\frac{2}{s^2 + \omega^2} (s^2 + \alpha) \hat{f} \right] (t) = \int_0^t \frac{2\sin(\omega(t - \tau))}{\omega} (f''(\tau) + \alpha f(\tau)) d\tau.$$

Each mode k then produces an absolutely convergent integral, and the spectral series acquires a classical meaning.

4.3.3 Final form in the time domain

We define two operators acting on a function f satisfying $f(0) = f'(0) = 0$:

$$\begin{aligned} [Nf](t) &= \frac{f(t)}{L} + \frac{2}{L} \sum_{k=1}^{\infty} \int_0^t \frac{\sin(\omega_{m,k}(t - \tau))}{\omega_{m,k}} (f''(\tau) + \alpha f(\tau)) d\tau, \\ [Df](t) &= \frac{f(t)}{L} + \frac{2}{L} \sum_{k=1}^{\infty} (-1)^k \int_0^t \frac{\sin(\omega_{m,k}(t - \tau))}{\omega_{m,k}} (f''(\tau) + \alpha f(\tau)) d\tau. \end{aligned} \quad (4.12)$$

The DtN flux then reads (matrix product)

$$[\Lambda]_0(t) = [Nf_0](t) - [Df_L](t), \quad [\Lambda]_L(t) = -[Df_0](t) + [Nf_L](t). \quad (4.13)$$

The following sections deal with the numerical implementation of (4.13) and with its validation.

4.4 Numerical validation

The purpose of this section is to *validate* formula (4.13) by confronting its evaluation with an independent reference solution. All the work described here was carried out in the notebook `Validation_Cylindre_PR_Massaró.ipynb`, whose main implementation choices and results are reported below.

4.4.1 Why the cylinder, and not the half-space?

The choice of the cylinder as a validation framework stems from its being a bounded domain. On the cylinder, the finite-difference grid covers exactly the physical domain $z \in (0, L)$, the spectrum is discrete – formula (4.13) being then a series that one truncates – and an explicit centered scheme suffices, under a simple CFL condition. On the half-space, one would instead have to truncate the domain at a fictitious boundary and equip it with an absorbing condition or a PML [7] in order to avoid spurious reflections contaminating the measured flux; the spectrum would be continuous and the singular kernel ($\text{Pf } r^{-3}$, supported on the light cone) would require a Hadamard finite-part quadrature on an unbounded domain. The extension to the half-space is therefore a perspective, but it presupposes a reliable PML and a carefully designed quadrature.

4.4.2 Validation strategy

We adopt the following strategy. For each fixed azimuthal mode m :

1. we choose two boundary data f_0 and f_L satisfying the vanishing initial conditions;
2. we compute the reference solution for the fluxes by direct resolution of the one-dimensional PDE, using high-resolution finite differences;
3. we independently evaluate the spectral formula (4.13) truncated at $k_{\max}$ terms;
4. we superimpose the two curves and measure the relative error.

If the formula is correct and the implementation faithful, the two curves must coincide to the accuracy expected from the truncations and discretizations involved.

4.4.3 Reference solution by finite differences

For each azimuthal mode m , the equation reduces to

$$\partial_t^2 u - \partial_z^2 u + \alpha u = 0, \quad u(0, t) = f_0(t), \quad u(L, t) = f_L(t), \quad u(\cdot, 0) = \partial_t u(\cdot, 0) = 0, \quad (4.14)$$

with $\alpha = (m/R)^2$. This equation, which is no longer the pure wave equation but the Klein-Gordon equation (mass term αu), is discretized on the grid $z_j = j\Delta z$ ($j = 0, \dots, N_z$, $\Delta z = L/N_z$) and $t_n = n\Delta t$ by the centered scheme

$$\frac{u_j^{n+1} - 2u_j^n + u_j^{n-1}}{\Delta t^2} = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta z^2} - \alpha u_j^n. \quad (4.15)$$

The scheme is solved explicitly, without inverting any system, and remains of order 2 in time as well as in space. Being centered, it does not dissipate energy and is therefore well suited to this type of hyperbolic equation.

Stability. For an explicit centered scheme applied to a hyperbolic equation, the von Neumann analysis imposes a condition relating the time step and the space step. For (4.14), it reads

$$\Delta t \leq \frac{\Delta z}{\sqrt{1 + \alpha \Delta z^2 / 4}}.$$

In practice, $\alpha \Delta z^2$ is very small compared with 1 (for instance, for $\Delta z \approx 1.5 \cdot 10^{-3}$ and $m = 100$, $\alpha \Delta z^2 \approx 0.023$), so that the denominator under the square root is nearly 1. The condition then reduces to *the classical Courant-Friedrichs-Lewy condition*

$$\Delta t \leq \Delta z.$$

In order to remain strictly inside the stability region, we choose in practice $\Delta t = 0.4 \Delta z$, that is, CFL = 0.4.

Computation of the boundary fluxes. Once u^{n+1} has been obtained, the fluxes are extracted by second-order one-sided formulas:

$$-\partial_z u(0, t) \approx \frac{3u_0 - 4u_1 + u_2}{2\Delta z}, \quad +\partial_z u(L, t) \approx \frac{3u_{N_z} - 4u_{N_z-1} + u_{N_z-2}}{2\Delta z}.$$

The implementation is given in Listing 4.1.

LISTING 4.1 – Reference solution by high-resolution finite differences

```

1 import numpy as np
2
3 def solve_wave_cylinder_mode(m, R, L, f0, fL, T, Nz=1000, CFL=0.4):
4     dz = L / Nz
5     dt = CFL * dz
6     Nt = int(np.ceil(T / dt))
7     dt = T / Nt # adjust so as to land exactly on T
8     alpha = (m / R) ** 2 # mass term (Klein-Gordon)
9
10    # Three time levels for the 3-point centered scheme
11    u_prev = np.zeros(Nz + 1)
12    u_curr = np.zeros(Nz + 1)
13    u_next = np.zeros(Nz + 1)
14    flux_0 = np.zeros(Nt + 1)
15    flux_L = np.zeros(Nt + 1)
16
17    for step in range(Nt):
18        t_new = (step + 1) * dt
19        u_curr[0] = f0(step * dt)
20        u_curr[-1] = fL(step * dt)
21
22        # Discrete 1D Laplacian (second-order centered difference)
23        d2z = (u_curr[2:] - 2*u_curr[1:-1] + u_curr[:-2]) / dz**2
24        # Scheme : u^{n+1} = 2u^n - u^{n-1} + dt^2 (u_zz - alpha u)
25        u_next[1:-1] = (2*u_curr[1:-1] - u_prev[1:-1]
26                        + dt**2 * (d2z - alpha*u_curr[1:-1]))
27        u_next[0] = f0(t_new)
28        u_next[-1] = fL(t_new)
29
30    # Normal derivatives by second-order one-sided formulas

```

```

31     d_plus = (-3*u_next[0] + 4*u_next[1] - u_next[2]) / (2*dz)
32     d_minus = ( 3*u_next[-1] - 4*u_next[-2] + u_next[-3]) / (2*dz)
33     flux_0[step + 1] = -d_plus          # outward normal at z=0
34     flux_L[step + 1] = +d_minus        # outward normal at z=L
35
36     u_prev, u_curr, u_next = u_curr, u_next, u_prev
37
38     return np.linspace(0, T, Nt + 1), flux_0, flux_L

```

With $L = 1.5$ and $N_z = 1000$, the spatial truncation error is of order $\Delta z^2 \sim 2.3 \times 10^{-6}$, that is, three orders of magnitude better than that of the spectral formula, which justifies the reference status granted to this solution.

4.4.4 Evaluation of the spectral formula

The numerical evaluation of (4.13) requires three choices.

Truncation of the series. The infinite sum over k must be truncated at some $k_{\max}$. To estimate the tail, we observe that $\omega_{m,k} \sim k\pi/L$ for large k , and that each spectral term is of the form

$$\frac{1}{\omega_{m,k}} \int_0^t \sin(\omega_{m,k}(t - \tau)) g(\tau) d\tau.$$

The factor $1/\omega_{m,k}$ is explicit, and the remaining oscillatory integral also decays like $1/\omega_{m,k}$ by the *Riemann-Lebesgue lemma* (the negative and positive contributions of $\sin(\omega\tau)$ cancel each other all the better as ω is large; one can be more precise by integration by parts if g is regular, which gives $O(1/\omega)$ for $g \in C^1$). The general term is therefore $O(1/\omega_{m,k}^2) = O(1/k^2)$, and the truncated tail satisfies

$$\sum_{k > k_{\max}} \frac{C}{k^2} = O\left(\frac{1}{k_{\max}}\right).$$

To aim at an accuracy of 10^{-2} , we choose $k_{\max} = 100$.

Remark 4.1 (Trade-off with temporal aliasing). *Increasing $k_{\max}$ does not reduce the error indefinitely. The kernels $\sin(\omega_{m,k}(t - \tau))$ oscillate at the frequency $\omega_{m,k}$. If $\omega_{m,k_{\max}} \Delta t$ becomes large compared with π , the quadrature no longer resolves these oscillations and the error increases again. There is thus an optimal value of $k_{\max}$ for a given temporal grid. The convergence curve of the notebook exhibits a minimum around $k_{\max} \approx 100$.*

Analytic second derivative. In order to avoid further numerical errors, f'' is supplied in analytic form. For the test datum $f(t) = t^2 e^{-at} \sin(wt)$, the second derivative is computed explicitly:

$$f''(t) = e^{-at} [A_S(t) \sin(wt) + A_C(t) \cos(wt)],$$

with $A_S(t) = -a(2t - at^2) + (2 - 2at - w^2 t^2)$ and $A_C(t) = -awt^2 + w(2t - at^2) + 2wt$.

Quadrature of the convolution. We denote here by $K = K_k$ the kernel of the current mode, defined by

$$K(t_i, \tau_j) = K_k(t_i, \tau_j) = \begin{cases} \frac{\sin(\omega_{m,k}(t_i - \tau_j))}{\omega_{m,k}} & \text{if } t_i \geq \tau_j, \\ 0 & \text{if } t_i < \tau_j, \end{cases} \quad \omega_{m,k} = \sqrt{\left(\frac{k\pi}{L}\right)^2 + \frac{m^2}{R^2}},$$

and by $g = f'' + \alpha f$ the source (arising from the derivative transfer). The distinction of cases reflects *causality*: the flux at time t_i depends only on the earlier values $\tau_j \leq t_i$. The integral $I(t_i) = \int_0^{t_i} K(t_i, \tau) g(\tau) d\tau$ is evaluated by the *composite trapezoidal rule*. On the grid $\tau_j = j \Delta t$, this rule reads

$$I(t_i) \approx \Delta t \left[\frac{1}{2} h(\tau_0) + h(\tau_1) + \cdots + h(\tau_{i-1}) + \frac{1}{2} h(\tau_i) \right], \quad h(\tau) = K(t_i, \tau) g(\tau).$$

We factor the quadrature as $I(t_i) = \sum_{j=0}^i W_{ij} K(t_i, \tau_j) g(\tau_j)$, where the weight matrix W is defined by

$$W_{ij} = \begin{cases} \Delta t/2 & \text{if } j = 0 \text{ or } j = i, \\ \Delta t & \text{if } 0 < j < i, \\ 0 & \text{if } j > i. \end{cases}$$

The matrix W is lower triangular, with half-weights at the endpoints. It is independent of the kernel K and of the mode k , and is therefore computed only once for the whole run of the program.

Vectorization. The discrete convolution then reduces to the matrix-vector product $[K * g]_i = ((W \odot K)g)_i$, where $\odot$ denotes the Hadamard product. In NumPy, this operation is written in two lines (`M = W * K` then `conv = M @ g`) and makes efficient use of optimized matrix computation routines, which makes the computation faster than an explicit loop.

Subsampling. The reference scheme uses a very fine temporal grid ($N_t \sim 10^4$ points for $\text{CFL} = 0.4$), imposed by the stability condition. At this resolution, the cost of the matrix convolution, $O(N_t^2)$ per mode k , becomes prohibitive (10^{10} operations in total for $k_{\max} = 100$).

We then exploit the fact that the fluxes Λf being sought are *smooth* functions of time. This regularity can be read directly off expression (4.13): each flux is the sum of the local term f/L and of convolutions $\int_0^t \sin(\omega_{m,k}(t - \tau)) / \omega_{m,k} \cdot g(\tau) d\tau$. Now convolution with a bounded kernel regularizes: if the datum f is of class C^∞ (which is the case for the chosen test profiles, $f(t) = t^2 e^{-at} \sin(wt)$), then $g = f'' + \alpha f$ is C^∞ as well, and so is each convolution. The resulting flux therefore exhibits neither discontinuity nor rapid oscillation in time: its representation on ~ 250 points over $[0, T]$ is largely sufficient to render it faithfully and to measure the error. We therefore perform an *a posteriori* subsampling (one point out of 40) of the reference solution computed at full resolution. The spectral formula is then evaluated on this coarse grid, which brings the total cost down to $\sim 6 \cdot 10^6$ operations. The subsampling does not degrade the accuracy of the reference: it simply selects a subset of already computed values, and in no case does it approximate them. The implementation is given in Listing 4.2.

LISTING 4.2 – Evaluation of the spectral formula (4.13)

```

1 def dtn_spectral_cylinder_mode(m, R, L, t_array,
2                               f0_vals, fL_vals,
3                               f0_dd_vals, fL_dd_vals,
4                               k_modes=100):
5     Nt = len(t_array)
6     dt = t_array[1] - t_array[0]
7     alpha = (m / R) ** 2
8
9     # Eigenfrequencies omega_{m,k} = sqrt((k pi / L)^2 + alpha)
10    ks = np.arange(1, k_modes + 1)
11    omegas = np.sqrt((ks * np.pi / L)**2 + alpha)
12
13    # Causal matrix of the time differences t_i - t_j (>= 0)

```

```

14 T_diff = t_array[:, None] - t_array[None, :]
15 mask = T_diff >= 0
16
17 # Trapezoidal quadrature matrix (lower triangular)
18 W = np.zeros((Nt, Nt))
19 for i in range(1, Nt):
20     W[i, 0] = dt / 2
21     if i > 1: W[i, 1:i] = dt
22     W[i, i] = dt / 2
23
24 # Convolved source g = f'' + alpha f (derivative transfer)
25 g0 = f0_dd_vals + alpha * f0_vals
26 gL = fL_dd_vals + alpha * fL_vals
27
28 # Initialization with the local term f/L of the operators N and D
29 N_f0, N_fL = f0_vals / L, fL_vals / L
30 D_f0, D_fL = f0_vals / L, fL_vals / L
31
32 # Spectral sum : each mode k adds one convolution
33 for k, omega in zip(ks, omegas):
34     sign = (-1) ** k
35     K = np.where(mask, np.sin(omega * T_diff) / omega, 0.0)
36     M = W * K
37     conv_0 = M @ g0
38     conv_L = M @ gL
39     N_f0 += (2.0 / L) * conv_0 # operator N (sign +)
40     N_fL += (2.0 / L) * conv_L
41     D_f0 += (2.0 / L) * sign * conv_0 # operator D (sign (-1)^k)
42     D_fL += (2.0 / L) * sign * conv_L
43
44 # Matrix recombination of the DtN flux at the two boundaries
45 lambda_0 = N_f0 - D_fL
46 lambda_L = -D_f0 + N_fL
47 return lambda_0, lambda_L

```

4.4.5 Comparison and error analysis

Choice of the test data. The validation uses two distinct temporal profiles on the two boundaries:

$$f_0(t) = t^2 e^{-0.5t} \sin(2.5t), \quad f_L(t) = t^2 e^{-0.3t} \sin(3.7t).$$

Both functions satisfy $f(0) = f'(0) = 0$. The choice of *different* profiles on Γ_0 and Γ_L is essential: if one took $f_0 = f_L$, the fluxes $[\Lambda]_0$ and $[\Lambda]_L$ would both be written $(N - D)f_0$ and only the combination $N - D$ would be tested. With distinct data, the operators N and D enter independently into each flux, and the full matrix system is exercised.

Sources of error. The global error comes from three contributions: the spectral truncation, in $O(1/k_{\max}) \sim 10^{-2}$ for $k_{\max} = 100$; the trapezoidal quadrature, in $O(\Delta t^2) \sim 10^{-4}$ for $N_t = 250$ on $[0, T]$ with $T = 3$; and the finite differences, in $O(\Delta z^2) \sim 10^{-6}$ for $\Delta z \approx 1.5 \cdot 10^{-3}$. The spectral truncation dominates.

Numerical results. For $R = 1$, $L = 1.5$, $T = 3$, $k_{\max} = 100$, $N_z = 1000$ and $N_t = 250$, the relative L^∞ errors obtained are reported in Table 4.1. The reference curves and those of the spectral formula coincide visually with an error of the order of one percent. This agreement simultaneously validates:

- the Mittag-Leffler expansion of $\mu_m \coth(\mu_m L)$ and $\mu_m / \sinh(\mu_m L)$;

- the derivative transfer, which makes each spectral term absolutely convergent;
- the *whole set* of eigenfrequencies $\omega_{m,k}^2 = (k\pi/L)^2 + (m/R)^2$ used: an error on the poles, even a minor one, would visibly degrade the superposition.

Mode m	Relative error at $z = 0$	Relative error at $z = L$
0	0.5 %	2.6 %
1	0.5 %	2.6 %
2	0.4 %	2.3 %
3	0.8 %	1.3 %

TABLE 4.1 – Relative L^∞ errors between the spectral formula (4.13) and the reference solution for the first azimuthal modes.

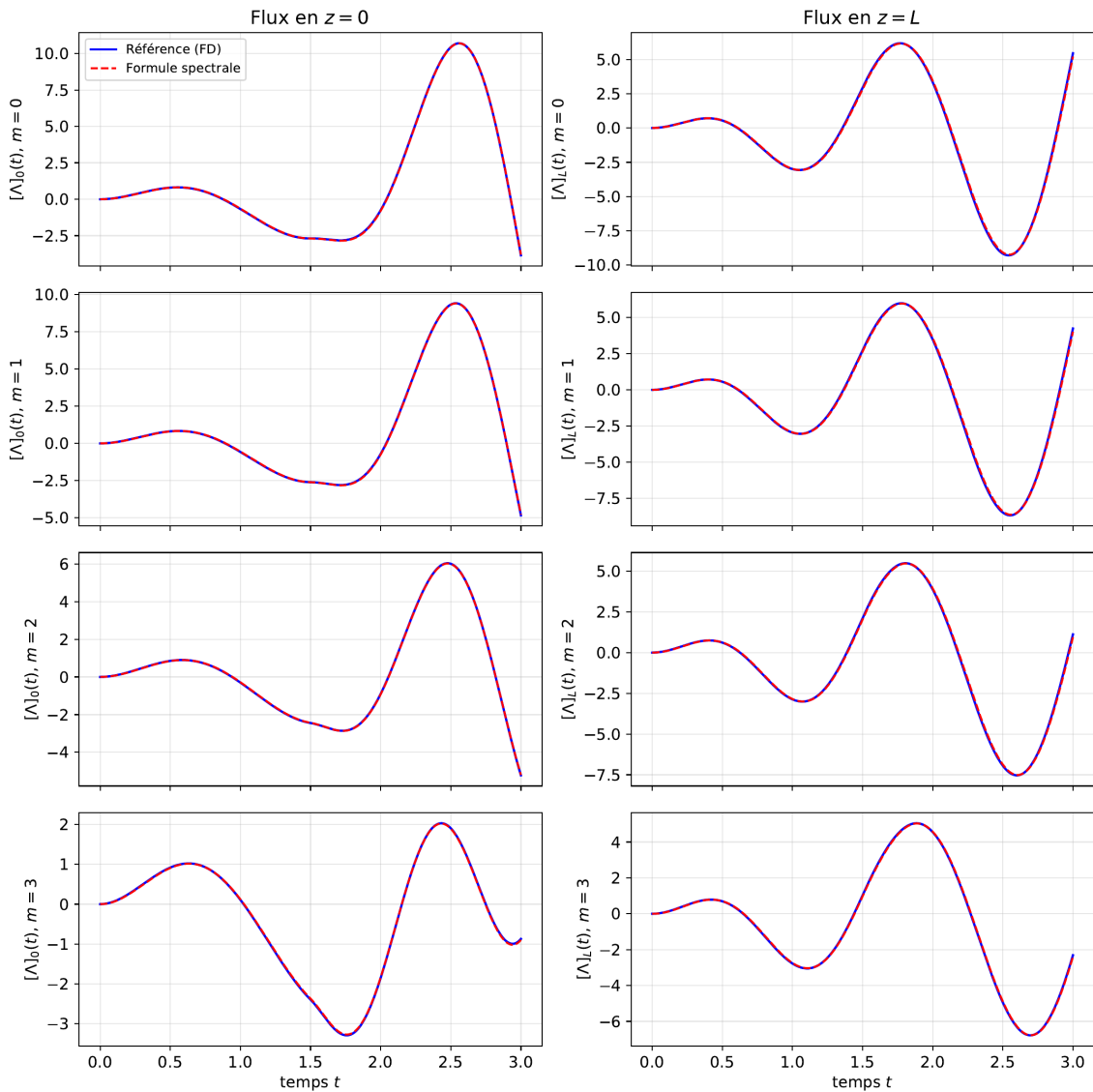


FIGURE 4.1 – Superposition of the reference solution by finite differences (solid blue line) and of the spectral formula (4.13) (dashed red line), for the modes $m = 0, 1, 2, 3$. The curves coincide visually to the L^∞ accuracy reported in Table 4.1.

The convergence study in $k_{\max}$ for $m = 2$ confirms the theoretical prediction: the error decreases

down to a minimum of about 0.4 % around $k_{\max} = 100$, and then increases again through aliasing, as illustrated in Figure 4.2.

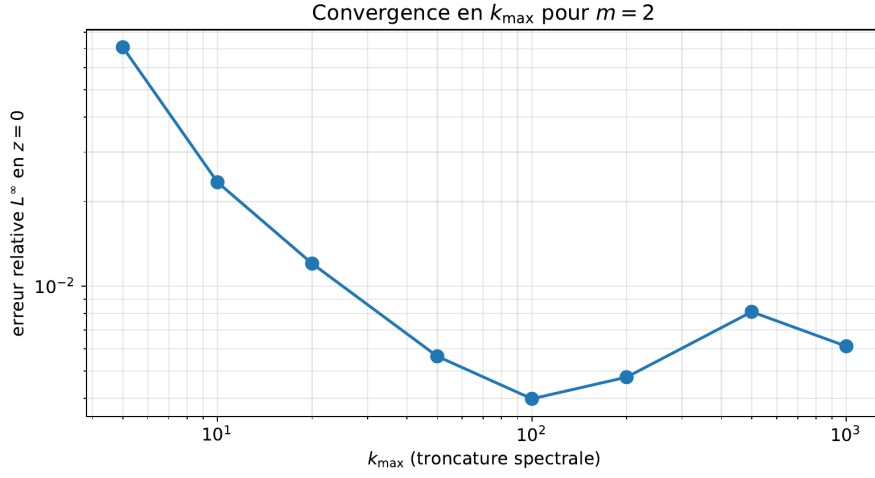


FIGURE 4.2 – Relative L^∞ error at $z = 0$ as a function of the spectral truncation $k_{\max}$, for the mode $m = 2$. The minimum around $k_{\max} \approx 100$ illustrates the trade-off between the decay of the truncated tail and temporal aliasing.

4.4.6 High-frequency limit: degradation at large modes

The preceding results concern the modes $m \in \{0, 1, 2, 3\}$, for which the formula attains an accuracy of the order of one percent. As m grows, this accuracy degrades rapidly:

$$\text{err}(m = 10) \approx 4 \%, \quad \text{err}(m = 40) \approx 18 \%, \quad \text{err}(m = 100) \approx 45 \%.$$

Figure 4.3 illustrates this degradation. Three combined mechanisms account for it.

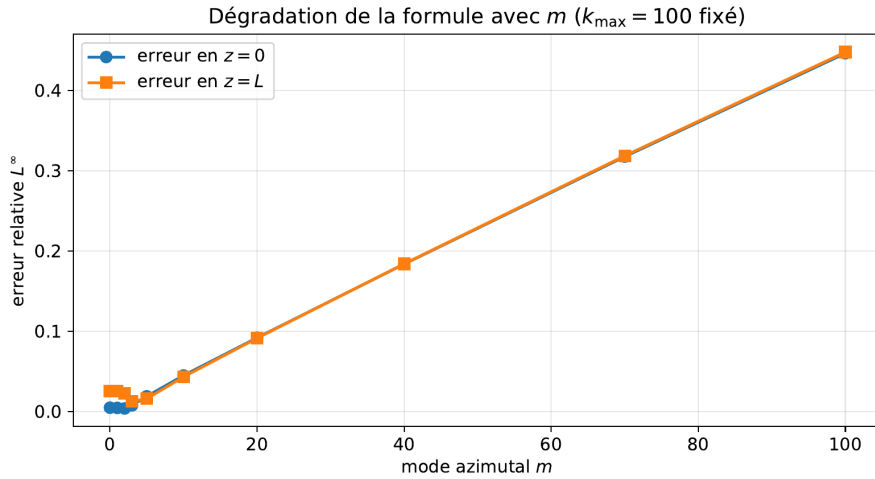


FIGURE 4.3 – Relative L^∞ error between the spectral formula and the reference solution, as a function of the azimuthal mode m , at fixed $k_{\max} = 100$. The accuracy degrades rapidly with m .

Clustering of the eigenfrequencies. The eigenfrequencies $\omega_{m,k} = \sqrt{(k\pi/L)^2 + \alpha}$, with $\alpha = (m/R)^2$, have a very different structure depending on whether m is small or large. For small m , α is negligible and the $\omega_{m,k}$ are essentially the harmonics $k\pi/L$, well separated. For large m , $\alpha = m^2/R^2$ dominates for the first values of k , and $\omega_{m,k} \approx \sqrt{\alpha} = m/R$: the first frequencies pile up almost

exactly at the threshold value m/R . For $m = 100$, $R = 1$, $L = 1.5$ for instance, $\omega_{100,1} \approx 100.022$, $\omega_{100,5} \approx 100.5$ and $\omega_{100,100} \approx 232$. The consequence is that the first terms of the series oscillate at almost the same frequency and that the “spectral budget” at fixed $k_{\max} = 100$ is poorly allocated: too many terms near m/R , not enough elsewhere, whence the loss of accuracy.

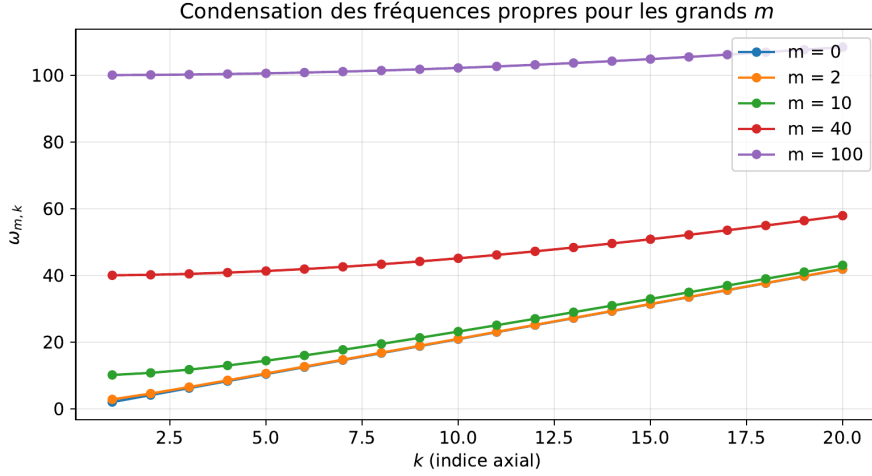


FIGURE 4.4 – Eigenfrequencies $\omega_{m,k}$ as a function of the axial index k , for various values of the azimuthal mode m . For large m , the first frequencies cluster around the common value m/R .

Amplification by the factor α . The source convolved in (4.13) is $g(\tau) = f''(\tau) + \alpha f(\tau)$, and for large m , $\alpha = (m/R)^2$ may greatly exceed $|f''|$: for the test datum and $m = 100$, the ratio $|\alpha f|/|f''|$ exceeds 10^4 . The trapezoidal quadrature error on g , intrinsically $O(\Delta t^2)$, is thus amplified and contaminates the result.

Limitation of the reference solution itself. For very large modes, the reference solution itself becomes inaccurate: the error of a second-order centered scheme is $\max |\partial_z^4 u| \Delta z^2$, and the fourth-order derivatives of a modal solution are of size $\sim m^4$. For $\Delta z = 1.5 \cdot 10^{-3}$ and $m = 100$, the factor $m^4 \Delta z^2 \approx 230$ becomes comparable to the values of the solution. A convergence study in N_z for $m = 100$ confirms this:

N_z	flux at $z = 0$ at $t = 1.5$
500	−59.07
1000	−60.28
2000	−60.62
4000	−60.71

The value has not stabilized with $N_z = 1000$; the “reference” still contains about 3 % of residual error, which contaminates the measurement of the discrepancy with the spectral formula.

Practical scope. The Fourier decomposition of a regular datum has coefficients f_m that decay rapidly with $|m|$: for $f \in C^p$, $|f_m| = O(|m|^{-p})$. Large modes $|m|$ therefore contribute little to the physical solution, and formula (4.13) is all the more usable in practice as the dominant modes are the low-frequency ones, where it is accurate. To treat very large modes rigorously, two directions are conceivable: adapting $k_{\max}$ to m (for instance $k_{\max} \sim Cm$) in order to preserve a uniform spectral coverage, or using a *Filon quadrature* [6] in place of the trapezoidal rule. The Filon quadrature is specifically designed for oscillatory integrals $\int f(\tau) e^{i\omega\tau} d\tau$: it interpolates the slow part $f(\tau)$

by a polynomial and integrates the product with the oscillatory kernel analytically, providing an accuracy independent of ω which circumvents precisely the aliasing problem encountered here.

4.4.7 Conclusion on the validation

The superposition of the curves and the error of the order of one percent obtained for the first modes validate at once the theoretical formula (4.13), the Laplace inversion strategy by derivative transfer, and the implementation.

5. Inverse problem: reconstruction of the geometric parameters

5.1 General setting

In the preceding chapters the domain was fixed and one characterized the associated operator Λ . We now reverse the question: given a (partial) measurement of Λ , can one reconstruct the geometric parameter of the domain?

This question is an instance of the *Calderón problem* [1] applied to simple geometries, where the only degree of freedom is a scalar parameter (radius or length). We first present the case of the disc, which serves as a reference, and then transpose the methods to the cylinder.

5.2 Reconstruction of the radius of a disc

5.2.1 Setting

We resume the setting of Chapter 3, now allowing the radius R to vary. Let $\Omega_R = \{|x| < R\} \subset \mathbb{R}^2$ and let Λ_R be the DtN operator associated with the wave equation on Ω_R . The modal computation leads, through the scaling $r \mapsto r/R$, to the symbol

$$\sigma_n(s; R) = s \frac{I'_n(sR)}{I_n(sR)}, \quad n \in \mathbb{Z}. \quad (5.1)$$

The Hadamard-Weierstrass factorization of Chapter 3 transposes into

$$\sigma_n(s; R) = \frac{|n|}{R} + \sum_{k=1}^{\infty} \frac{2s^2/R}{s^2 + (j_{n,k}/R)^2}, \quad (5.2)$$

where $(j_{n,k})_k$ are the positive zeros of J_n . The presence of the factor $1/R$ in the numerator is imposed by the differentiation: writing $z = sR$, the identity $I'_n(z)/I_n(z) = n/z + \sum_k 2z/(z^2 + j_{n,k}^2)$ gives

$$\sigma_n(s; R) = s \frac{I'_n(sR)}{I_n(sR)} = s \left[\frac{n}{sR} + \sum_k \frac{2(sR)}{(sR)^2 + j_{n,k}^2} \right] = \frac{|n|}{R} + \sum_k \frac{2s^2/R}{s^2 + (j_{n,k}/R)^2},$$

which ensures dimensional consistency (σ_n has the dimension of an inverse length, as expected of a DtN operator). The eigenfrequencies are the imaginary poles $s = \pm i j_{n,k}/R$, dilated by the factor $1/R$ relative to the unit disc.

5.2.2 Two reconstruction methods

Method A — Static limit. For every $n \neq 0$, $\lim_{s \rightarrow 0} \sigma_n(s; R) = |n|/R$, whence

$$R = \frac{|n|}{\sigma_n(0^+; R)}.$$

This method is conceptually the simplest, but it requires extrapolating to zero frequency, which is numerically delicate.

Method B — Resonances. The poles of $\sigma_n(\cdot; R)$ are $s = \pm i j_{n,k}/R$. The k -th eigenfrequency of the mode n therefore equals $\omega_{n,k} = j_{n,k}/R$. One deduces

$$R = \frac{j_{n,k}}{\omega_{n,k}}.$$

In practice, one excites a single mode, computes the temporal Fourier transform of the response, and reads off the first eigenfrequency. This method is robust, since resonances are stable and easily identifiable quantities.

5.3 Reconstruction of the length of a cylinder

5.3.1 Setting

We consider the cylinder $\Omega = S_R^1 \times (0, L)$ of *known* radius R , and we seek to reconstruct the length L from a measurement of Λ . By Fourier-Laplace decomposition (*cf.* Chapter 4), the matrix symbol reads

$$\hat{\Lambda}_m(s) = \begin{pmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{pmatrix} = \begin{pmatrix} \mu_m \coth(\mu_m L) & -\frac{\mu_m}{\sinh(\mu_m L)} \\ -\frac{\mu_m}{\sinh(\mu_m L)} & \mu_m \coth(\mu_m L) \end{pmatrix}, \quad \mu_m = \sqrt{s^2 + \frac{m^2}{R^2}}. \quad (5.3)$$

5.3.2 Three inversion methods

Three reconstruction strategies for L follow directly from the form of the symbol.

Method A — Via the diagonal coefficient. Since $\lambda_{11}/\mu_m = \coth(\mu_m L)$ and $\coth$ is strictly decreasing on $\mathbb{R}_+^*$, the inversion is explicit:

$$L = \frac{1}{\mu_m} \operatorname{arccoth}\left(\frac{\lambda_{11}}{\mu_m}\right) = \frac{1}{2\mu_m} \ln\left(\frac{\lambda_{11} + \mu_m}{\lambda_{11} - \mu_m}\right). \quad (5.4)$$

Method B — Via the ratio. The quotient of the coefficients eliminates μ_m :

$$-\frac{\lambda_{12}}{\lambda_{11}} = \frac{1}{\cosh(\mu_m L)} \implies L = \frac{1}{\mu_m} \operatorname{arccosh}\left(-\frac{\lambda_{11}}{\lambda_{12}}\right).$$

The advantage of this formulation is that it depends only on a *ratio*, and therefore remains robust to a multiplicative error factor on the measurements.

Method C — High-frequency asymptotics. As $\mu_m \rightarrow \infty$, the off-diagonal coefficient admits the expansion

$$-\lambda_{12} \sim 2\mu_m e^{-\mu_m L},$$

whence

$$L = \lim_{\mu_m \rightarrow \infty} \frac{1}{\mu_m} \ln\left(\frac{2\mu_m}{-\lambda_{12}}\right). \quad (5.5)$$

5.3.3 Identifiability

The strict monotonicity of $\coth$ ensures the uniqueness of the solution L given by Method A: a single measurement (m, s) with $\mu_m > 0$ therefore suffices to reconstruct L . The other methods

allow a cross-check: a discrepancy between the estimates obtained through three different channels (diagonal, ratio, asymptotic) is a good indicator of measurement errors or numerical noise.

6. Conclusion and perspectives

This research project allowed me to address the explicit characterization of a non-trivial pseudodifferential operator, the Dirichlet-to-Neumann operator for the wave equation, in four situations of differing complexity.

On the unperturbed half-space, two independent computations of the kernel were carried out: one by Bessel/Hankel tables and finite part, the other by computing the distributional Laplacian. Their explicit reconciliation, and the identification of the non-local part with the expected symbol $\sqrt{-\Delta_{x'}}$, remain to be completed.

The treatment of the half-space perturbed by a potential via the Born approximation showed how the Lippmann-Schwinger integral formulation, combined with a relative change of variables and with the Poisson formula for the wave equation, leads to an explicit expression of Λ_q as a linear correction of Λ_0 . The derivation of the Poisson formula gives back the DtN kernel and provides a consistency cross-check.

On the unit disc, the bounded geometry gives rise to resonances in the form of imaginary poles at the zeros of the Bessel functions. The Hadamard-Weierstrass factorization together with the derivative transfer yield an absolutely convergent spectral expression, which serves as the basis for the inverse problem.

On the finite hollow cylinder, the matrix nature of the symbol introduces a non-trivial coupling between the two boundaries. The Laplace inversion proceeds through a Mittag-Leffler expansion combined with a derivative transfer that makes each term absolutely convergent. The numerical validation carried out in the notebook confirmed the theoretical formula with an accuracy of the order of one percent on the first modes, and brought to light three mechanisms of degradation at large azimuthal modes.

On the inverse problem, the DtN operator encodes geometric parameters: the radius in the case of the disc, the length in the case of the cylinder. In each case, several methods (static, resonances, high-frequency asymptotics) coexist, and their applicability depends on the model.

Perspectives. Three directions would have deserved further investigation: the complete reconciliation of the two half-space computations and their formalization through microlocal analysis [2], which would unify the three cases encountered and clarify the status of the part $\sqrt{-\Delta_{x'}}$; the extension of the perturbed treatment by Born approximation to the cylinder, via the explicit Green's function; and the numerical validation on the half-space, which presupposes a reliable absorbing condition [7] and a carefully designed finite-part quadrature. Beyond the technical aspects, this experience gave me an introduction to the world of research in PDE analysis, and confirmed my interest in this balance between theoretical formulation, explicit computation and numerical validation.

A. Validation notebook data

Physical parameters. $R = 1$, $L = 1.5$, $T = 3$.

Numerical parameters. $N_z = 1000$, $\text{CFL} = 0.4$, $N_t = 250$ after subsampling, $k_{\max} = 100$.

Boundary data.

$$f_0(t) = t^2 e^{-0.5t} \sin(2.5t), \quad f_L(t) = t^2 e^{-0.3t} \sin(3.7t).$$

Source code. The whole code can be consulted in the notebook `Validation_Cylindre_PR_Massaro.ipynb` provided as an electronic appendix.

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